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# Step 1: Import required libraries
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import pandas as pd

# Step 2: Load the built-in dataset
data = load_breast_cancer()

# Step 3: Create DataFrame
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target)

# Step 4: Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Step 5: Create a function for Random Forest
def random_forest(X_train, X_test, y_train, y_test):

    # Step 6: Initialize the Random Forest model
    model = RandomForestClassifier(
        n_estimators=100,
        random_state=42
    )

    # Step 7: Train the model
    model.fit(X_train, y_train)

    # Step 8: Make predictions
    y_pred = model.predict(X_test)

    # Step 9: Calculate Accuracy
    accuracy = accuracy_score(y_test, y_pred)
    print("Accuracy:", accuracy)

    # Step 10: Display Confusion Matrix
    print("\nConfusion Matrix:")
    print(confusion_matrix(y_test, y_pred))

    # Step 11: Display Classification Report
    print("\nClassification Report:")
    print(classification_report(y_test, y_pred))

    # Step 12: Compare Actual vs Predicted
    comparison = pd.DataFrame({
        "Actual": y_test.values,
        "Predicted": y_pred
    })

    print("\nActual vs Predicted:")
    print(comparison.head(10))

# Step 13: Call the function
random_forest(X_train, X_test, y_train, y_test)
```

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Accuracy: 0.9649122807017544

Confusion Matrix:
[[40  3]
 [ 1 70]]

Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.93      0.95        43
           1       0.96      0.99      0.97        71

    accuracy                           0.96       114
   macro avg       0.97      0.96      0.96       114
weighted avg       0.97      0.96      0.96       114
```

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Actual vs Predicted:
   Actual  Predicted
0       1          1
1       0          0
2       0          0
3       1          1
4       1          1
5       0          0
6       0          0
7       0          0
8       1          0
9       1          1
```